

8(a)	newton per ampere per metre	M1
	where current/wire is perpendicular to magnetic field	A1
8(b)(i)	$F = BIL \sin \theta$	C1
	$B = 1.0 / (5.0 \times 0.060 \times \sin 50^\circ)$ $= 4.4 \text{ mT}$	A1
8(b)(ii)	(from Fleming's left-hand rule) force on wire is upwards, so reading decreases	B1
8(b)(iii)	frame will rotate (so that PQ becomes perpendicular to the field)	B1

Question	Answer	Marks
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